

WENATCHEE PANGBORN, WA, USA

WMO: 727825

Lat: 47.398N

Lon: 120.201W

Elev: 1229

StdP: 14.05

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94239

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.9	14.7	1.6	6.2	13.9	6.7	8.1	18.3	21.5	39.9	18.2	39.6	2.8	290	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.7	96.2	64.6	92.8	63.9	89.4	62.8	67.1	91.0	65.5	88.4	64.0	85.9	7.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	74.0	75.6	55.6	68.8	74.8	54.2	65.4	74.1	32.1	91.0	30.8	88.2	29.7	85.9	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.1	20.5	18.1	DB	5.5	101.5	6.8	3.2	0.6	103.8	-3.3	105.7	-7.1	107.4	-12.0	109.8	(4)
(5)			WB	4.6	69.2	6.5	1.8	-0.1	70.4	-4.0	71.5	-7.6	72.5	-12.4	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	29.5	34.6	43.5	51.4	60.3	66.9	75.6	74.4	65.1	51.1	38.3	29.4
	DBStd	17.49	7.51	7.32	6.44	6.08	7.22	6.91	6.45	6.03	6.63	6.63	7.32	7.27
	HDD50	2393	635	431	214	52	5	0	0	0	1	62	355	638
	HDD65	5689	1100	851	667	409	183	55	4	5	79	431	802	1103
	CDD50	3045	0	0	13	95	325	507	794	756	454	98	3	0
	CDD65	863	0	0	0	1	39	111	332	296	82	2	0	0
	CDH74	9330	0	0	0	33	457	1204	3790	3053	776	17	0	0
	CDH80	4174	0	0	0	3	139	479	1903	1398	249	3	0	0
Wind	WSAvg	6.8	3.3	4.3	6.6	8.7	9.2	10.1	9.9	9.0	7.4	5.9	4.2	3.3
Precipitation	PrecAvg	8.3	1.1	0.8	0.6	0.6	0.6	0.6	0.3	0.4	0.3	0.4	1.1	1.3
	PrecMax	14.3	2.4	2.3	2.3	3.3	2.0	1.9	1.1	2.0	2.3	1.7	3.6	2.8
	PrecMin	4.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	2.0	0.6	0.6	0.5	0.6	0.6	0.5	0.3	0.5	0.5	0.4	0.7	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	56.1	67.3	78.6	88.8	97.5	101.4	98.8	91.9	75.6	58.7	48.6
		MCWB	41.3	45.3	51.1	56.4	60.6	65.0	65.8	66.4	64.2	57.5	48.9	40.9
	2%	DB	44.5	50.6	61.9	72.4	84.1	90.2	97.3	95.0	86.4	69.7	54.4	44.4
		MCWB	38.5	42.1	47.8	53.7	59.4	62.1	64.8	64.7	61.9	54.6	46.5	38.9
	5%	DB	40.6	47.4	57.8	67.6	79.2	85.4	93.9	91.4	82.2	65.6	51.1	40.7
		MCWB	36.3	40.2	45.3	50.9	57.5	60.3	63.9	64.2	60.7	52.2	44.8	36.5
	10%	DB	37.2	44.2	54.5	63.4	74.4	81.0	90.4	88.0	78.4	62.5	48.2	37.6
		MCWB	34.6	38.2	43.6	48.5	55.4	59.3	63.2	62.7	58.6	50.6	43.2	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.0	46.5	52.2	58.3	63.2	66.8	69.4	69.1	65.9	59.1	51.1	42.1
		MCDB	48.2	54.0	64.8	75.6	84.0	91.9	94.2	92.6	86.6	72.0	56.4	46.8
	2%	WB	39.1	43.3	48.8	54.6	60.4	63.9	67.4	67.0	63.7	55.9	48.3	39.3
		MCDB	43.4	49.5	60.2	69.9	80.3	86.4	91.7	90.3	83.5	66.1	52.4	42.9
	5%	WB	36.9	40.8	46.4	51.9	58.5	61.9	65.7	65.3	61.6	53.5	46.1	37.1
		MCDB	39.8	46.2	56.0	66.4	76.5	83.0	89.5	87.4	79.8	63.6	50.2	39.8
	10%	WB	35.1	38.8	44.3	49.6	56.5	59.9	64.2	63.8	59.6	51.5	43.7	35.2
		MCDB	37.1	43.3	53.0	62.2	72.2	78.8	87.0	85.1	75.8	60.7	47.7	37.4
Mean Daily Temperature Range	5% DB	MDBR	8.3	12.7	17.2	19.8	21.5	22.0	24.7	23.9	22.5	18.1	11.7	7.9
		MCDBR	13.9	17.4	21.9	25.5	27.8	27.4	28.5	27.7	27.2	22.0	15.5	13.2
		MCWBR	9.5	10.6	11.9	12.2	11.1	9.6	9.0	9.6	11.3	11.1	9.5	9.3
	5% WB	MCDBR	12.2	16.0	20.6	24.5	25.3	25.6	26.8	26.5	26.0	20.2	13.4	11.6
		MCWBR	8.6	10.3	11.9	12.5	11.0	10.5	9.8	10.0	11.5	11.0	8.9	8.3
Clear-Sky Solar Irradiance	taub	0.285	0.289	0.302	0.322	0.336	0.341	0.340	0.335	0.328	0.318	0.297	0.283	
	taud	2.430	2.504	2.530	2.461	2.455	2.463	2.474	2.485	2.496	2.515	2.521	2.460	
	Ebn at Noon	256	283	295	297	294	292	291	288	281	267	251	243	
	Edh at Noon	21	24	28	33	34	34	33	31	28	24	19	18	
All-Sky Solar Radiation	RadAvg	348	660	1020	1512	1858	2077	2306	1940	1407	837	423	278	
	RadStd	44	81	92	115	134	141	123	115	94	87	51	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+68

Nomenclature: See separate page